

BAXTER FABRIC

Federated Analytics & Business Resource Intelligence Core

Solution Architecture | Implementation Roadmap | PMO Structure

Enterprise Intelligence at Open-Source Economics

Palantir-grade semantic capability | Siemens AAS acceleration | Four-domain governance

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Architecture Framework: Requirements-Driven Semantic Gateway (RDSG v2.0)

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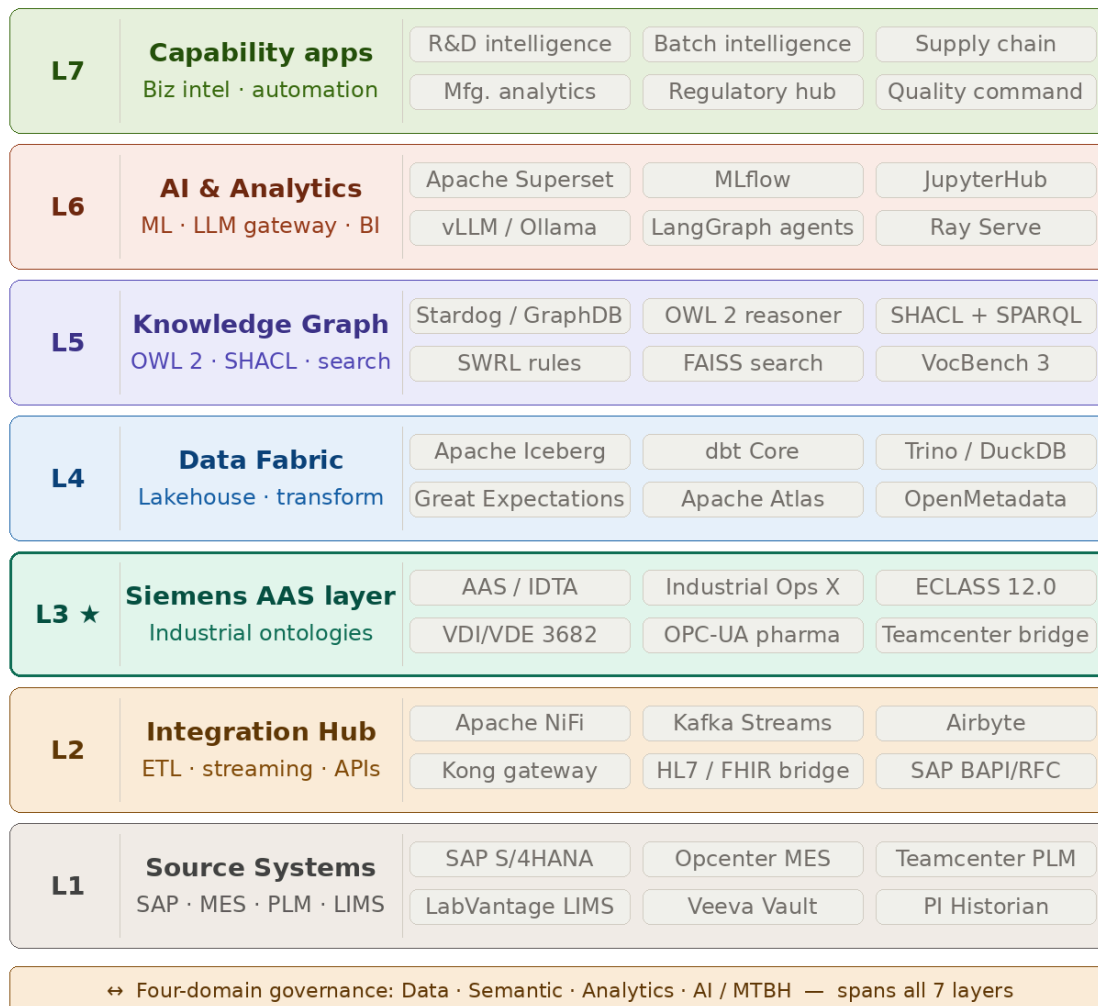
1. Executive Summary

Baxter Healthcare operates one of the most complex enterprise technology landscapes in the medical device and pharmaceutical industry — spanning four major business lines (Renal Care, Medication Delivery, Hospital Products, Advanced Surgery), dozens of global manufacturing facilities, and a heterogeneous stack of ERP, MES, PLM, LIMS, QMS, and clinical systems accumulated through decades of organic growth and M&A activity.

This document presents the Baxter FABRIC Solution Architecture — a seven-layer, semantically governed enterprise intelligence platform providing Palantir Foundry-equivalent capabilities at 75-80% lower total cost of ownership. FABRIC is designed to accelerate R&D knowledge synthesis, streamline manufacturing and supply chain analytics, and establish a holistic four-domain governance framework spanning data, semantic, analytics, and AI governance.

Baxter FABRIC — Seven-Layer Architecture

Federated Analytics & Business Resource Intelligence Core
Open-source · Siemens AAS-accelerated · MTBH-governed · Four-domain governance



★ Siemens AAS layer (L3) eliminates 6–12 months of custom ontology development
Value flow: L1 source systems → L7 capability applications

The architecture's central accelerator is the integration of Siemens' Asset Administration Shell (AAS) standard and Industrial Operations X (IOX) platform at the semantic harmonization layer — eliminating 6-12 months of custom ontology development by leveraging pre-built IDTA industrial ontologies, Siemens OPC-UA companion specifications for pharmaceutical manufacturing, and the ECLASS 12.0 product classification taxonomy as a seed ontology for Baxter's domain model. The AAS provides what Palantir calls its "Ontology" — but on open international standards, without proprietary lock-in.

Key Value Metrics

Metric	Palantir Foundry	Baxter FABRIC	Delta
Platform TCO (annual, steady state)	\$8M – \$15M	\$750K – \$1.5M	75–80% reduction
Time to first R&D insight	12–18 months	8–12 weeks	~6x faster
Year 1 implementation investment	\$3M – \$6M (SI fees)	\$600K – \$1.1M	\$2–5M savings
Ontology standard	Proprietary (Foundry OLS)	Open (IEC 63278 AAS / OWL 2)	Portable; no lock-in
Vendor lock-in risk	High	Low (open standards throughout)	Architectural freedom
Regulatory AI compliance	Requires custom build	Native (MTBH / EU AI Act ready)	Reduced compliance risk
Governance domains	2 (data + analytics)	4 (+ semantic + AI/MTBH)	AI governance built in

Strategic Rationale

FABRIC is not a cost reduction exercise alone — it is a strategic repositioning of Baxter's data and AI capability on an open, standards-based foundation that avoids the architectural debt of proprietary platforms. The four governance domains are designed to position Baxter ahead of the FDA's evolving AI/ML guidance for pharmaceutical manufacturing and the EU AI Act's requirements for high-risk AI in healthcare.

The R&D entry point is deliberate. R&D holds the highest density of siloed, high-value data — formulation knowledge, stability findings, analytical method repositories — that has never been harmonized across Baxter's business lines. Unlocking this knowledge through FABRIC creates immediate measurable value while building the semantic infrastructure that all subsequent phases depend upon. Every semantic model built for R&D extends directly to manufacturing and supply chain without rework.

2. Architecture Overview

The Baxter FABRIC architecture is organized into seven functional layers, each with a distinct responsibility and defined interface contract with adjacent layers. The architecture follows a "semantic-first" design philosophy — every entity, relationship, and metric has a formal ontological definition before it is exposed to analytics or AI consumers. A cross-cutting four-domain governance framework operates above all seven layers as an ambient property of the entire platform.

Layer	Name	Primary Function	Key Technologies	Phase
7	Capability Applications	Business-facing intelligence surfaces and process automation	Superset, LangGraph agents, workflow automation	1–3
6	AI & Analytics	ML lifecycle, generative AI gateway, governed analytics	MLflow, JupyterHub, vLLM, FLAML, Ray Serve, Grafana	1–3
5	Knowledge Graph	Semantic store, OWL reasoning, cross-domain SPARQL query	Stardog/GraphDB, OWL 2, SHACL, SPARQL 1.1, SWRL	1–3
4	Data Fabric	Physical storage, SQL transformation, quality, lineage	Apache Iceberg, dbt Core, Trino, Great Expectations, Atlas	1–3
3 ★	Semantic Harmonization (Siemens)	Universal entity model, AAS, industrial ontologies	Siemens AAS, IOX, IDTA, ECLASS 12.0, VDI/VDE 3682	1–2
2	Integration Hub	Data movement, protocol bridging, event streaming	Apache NiFi, Kafka, Airbyte, IOX OPC-UA, Kong	1–2
1	Source Systems	Systems of record — all BLs, product lines, facilities	SAP, Opcenter MES, Teamcenter, LIMS, Veeva, PI Historian	1–2

PRINCIPLE

All seven layers are governed by the cross-cutting Four-Domain Governance Framework (data, semantic, analytics, AI/MTBH). Governance is not a layer — it is an ambient property of the entire architecture, enforced at every layer boundary.

3. Layer Architecture Breakouts

Layer 1: Source Systems

3.1.1 Purpose & Scope

The source layer represents all authoritative systems of record across Baxter's R&D, manufacturing, and supply chain operations. These are the origin systems for all data flowing into FABRIC. Critically, FABRIC operates in read-only integration mode — no source system is modified, preserving GxP system validation status and eliminating change management risk at the ERP, MES, and LIMS level. All source connectivity is governed by formal data access agreements and monitored via the integration hub's audit log.

3.1.2 Component Inventory

Component	Category	Description and Baxter Application
SAP S/4HANA	ERP	Enterprise resource planning covering finance, procurement, production orders, inventory, and supplier master data. Primary authority for BOM, cost center, and supply chain master data across all business lines. Multiple instances per BL — unified via FABRIC's integration layer.
Siemens Opcenter MES	MES/MOM	Manufacturing execution and operations management — electronic batch records (eBR), WIP tracking, material genealogy, equipment management, and ISA-95 order management. The natural entry point for Siemens IOX integration.
Siemens Teamcenter PLM	PLM	Product lifecycle management — product structure, design history file (DHF), engineering change orders (ECO/ECR), and design documentation. Primary source for multi-level BOM and regulatory change management linkage.
LabVantage / IDBS LIMS	LIMS	Laboratory information management — analytical test results, sample chain of custody, specification management, stability study data, and method validation records. Per-BL configurations with divergent schemas.
Veeva Vault QMS/RIM	Quality/Regulatory	Quality management (CAPA, deviations, change control, SOPs) and regulatory information management (submissions, product registrations, labeling). Authority for all regulated document content.
OSIsoft PI Historian	Process Data	Time-series operational data from manufacturing — temperature, pressure, pH, flow, OEE metrics, and environmental monitoring. Ingested via Kafka for real-time analytics and process deviation detection.
SCADA / DCS Systems	Control Systems	Supervisory control and data acquisition systems across manufacturing facilities. Real-time process control data surfaced via OPC-UA adapters through Siemens Industrial Operations X — no custom PLC integration required.
ELN (Benchling / IDBS)	R&D	Electronic lab notebooks — formulation records, experimental design, reaction conditions, and analytical observations across R&D business lines. Multiple ELN instances with incompatible schemas — harmonized via FABRIC Layer 3.
OSIsoft PI AF	Asset Framework	Asset framework for contextualizing PI Historian data with equipment hierarchy and process element definitions. Feeds directly into AAS submodel structure in Layer 3.

Component	Category	Description and Baxter Application
CRO / Clinical Data	Clinical	Contract research organization data feeds, clinical trial repositories, and bioanalytical data for product pipeline decisions. Ingested via HL7/FHIR bridge for applicable product lines.

3.1.3 Baxter-Specific Considerations

Baxter operates 40+ distinct systems across four major business lines and dozens of global manufacturing facilities. Each business line has evolved its system stack independently, resulting in multiple ERP instances, independent LIMS configurations, separate quality systems, and divergent ELN deployments. FABRIC treats this heterogeneity as a feature, not a problem — the integration and semantic harmonization layers are specifically designed to work above this diversity without forcing source system consolidation.

3.1.4 Key Architecture Decisions

- Read-only integration in Phase 1 and Phase 2 — write-back to source systems deferred to Phase 3 and only for non-GxP applications
- FABRIC Canonical Identifiers (FCIDs) created at integration layer; source system keys preserved as attributes for backward traceability
- Refresh cadence driven by source SLAs: near-real-time for Historian/SCADA, hourly for MES, daily for ERP/LIMS/Veeva
- Phase 1 prioritizes R&D systems (ELN, LIMS, SAP R&D objects); Phase 2 extends to manufacturing and supply chain
- GxP system integrations require Computer System Assurance (CSA) assessment per FDA guidance before activation

Layer 2: Integration Hub

3.2.1 Purpose & Scope

The integration hub provides all data movement, protocol translation, and event streaming capabilities that connect source systems to the semantic harmonization and data fabric layers. It handles schema mapping, protocol bridging (SAP BAPI/RFC, OPC-UA, HL7/FHIR, REST, MQTT), batch ETL, and real-time event streaming. Every data movement in the integration hub is logged, auditable, and replayable — providing a foundational audit trail that supports both operational troubleshooting and regulatory evidence collection. This layer is designed for fault tolerance and idempotent operation: any flow can be replayed from the Kafka retention window without downstream duplication.

3.2.2 Component Inventory

Component	Category	Description and Baxter Application
Apache NiFi	Data Orchestration	Visual flow-based data pipeline engine for batch ETL. Native SAP BAPI/RFC support via GetSAPTable processor. Built-in data provenance captures full lineage of every record movement — every NiFi flow graph is a lineage artifact readable by Apache Atlas.
Apache Kafka / Kafka Streams	Event Streaming	Distributed event streaming for real-time data sources (Historian, SCADA, IoT, MES events). Topic-per-source-domain design with configurable retention (30 days default). Enables event-driven architecture for manufacturing alerting, process deviation detection, and near-real-time batch intelligence.
Airbyte (Open Source)	Connector Library	300+ pre-built connectors for SaaS and database sources. Manages change data capture (CDC) for relational source systems. Supplements NiFi for standard vendor connectors — avoids custom connector development for common integration patterns.
Siemens IOX OPC-UA Adapter	Industrial Protocol	Siemens Industrial Operations X provides production-grade OPC-UA adapter for SCADA, DCS, PLCs, and SIMATIC automation assets. Eliminates custom protocol development for industrial data ingestion — a primary accelerator for the manufacturing domain integration.
Kong API Gateway	API Management	Manages REST and GraphQL API ingestion from third-party sources (CROs, CMOs, suppliers, regulatory data feeds). Provides rate limiting, mutual TLS authentication, schema validation, and comprehensive audit logging for all external data consumers.
HL7 / FHIR Bridge	Healthcare Standards	For Baxter product lines with clinical data requirements — HL7 v2 and FHIR R4 message transformation to FABRIC canonical format. Supports pharmacovigilance data flows and clinical outcome data feeds for R&D pipeline decisions.
MuleSoft / Apache Camel	Enterprise Integration	For complex SAP integration patterns requiring transactional guarantees (RFC, BAPI, IDoc). Used where NiFi's eventual-consistency model is insufficient — primarily for SAP master data synchronization and order-critical transactions.

3.2.3 Baxter-Specific Considerations

The integration hub must serve Baxter's diverse technology stack across global facilities without requiring any source system modification. Key challenges include: multi-instance SAP (separate ERP instances per BL with varying release levels), heterogeneous MES configurations, HL7/FHIR requirements for clinical product lines, and CMO/supplier data feeds with varying formats and reliability. The Siemens IOX

OPC-UA adapter is particularly valuable — it converts Baxter's entire manufacturing floor data collection from a custom integration problem into a standard configuration exercise.

3.2.4 Key Architecture Decisions

- NiFi as primary batch orchestration; Kafka for all real-time streams; Airbyte for standard SaaS connectors
- All integration flows are idempotent — safe for replay from Kafka retention window (30 days default)
- Integration hub operates in DMZ network segment — no direct database connections to production source systems
- Siemens IOX handles all industrial protocol translation — custom OPC-UA/MQTT adapter development eliminated
- Message schema registry (Confluent Schema Registry compatible) enforces schema contracts at Kafka topic level

Layer 3: Semantic Harmonization — Siemens AAS

3.3.1 Purpose & Scope

The semantic harmonization layer is the most architecturally distinctive component of FABRIC and the principal differentiator from conventional data platforms. It uses the Siemens Asset Administration Shell (AAS) standard (IEC 63278 / IDTA) and associated IDTA industrial ontologies as a universal entity model — giving every asset, batch, material, supplier, process, and product a typed, semantically rich digital identity before any analytics or knowledge graph layer sees it. This layer eliminates 6-12 months of custom ontology development by leveraging Siemens' pre-built industrial semantic models, and positions Baxter's architecture on open international standards rather than proprietary constructs. The AAS model replaces Pantir Foundry's proprietary ontology objects with an open, standards-based, and reusable equivalent.

3.3.2 Component Inventory

Component	Category	Description and Baxter Application
Asset Administration Shell (AAS)	Universal Entity Model	IDTA standard IEC 63278. Every Baxter asset — equipment, batch, material, product, supplier, facility — is represented as an AAS with a unique global identifier and typed submodels. The AAS Registry maintains the authoritative index of all shells. AAS serialization in JSON-LD for knowledge graph ingest.
Siemens Industrial Operations X (IOX)	IIoT Semantic Platform	Cloud-based industrial IoT platform providing semantic context for operational data. IOX maps PLC/SCADA signals to AAS submodel attributes — transforming raw sensor values into semantically typed measurements with units, ranges, and quality annotations.
IDTA Submodel Templates	Pre-built Semantics	IDTA-maintained submodel templates available immediately: NameplateV2, TechnicalData, HandoverDocumentation, ContactInformation, ProductCarbonFootprint, SoftwareNameplate, DigitalNameplate. Baxter extends with pharma-specific submodels (BatchRecord, StabilityProfile, SupplierQualification, RegulatoryDossierLink).
ECLASS 12.0	Product Classification	Industry-standard product and service classification taxonomy with 49,000+ classes and 17,000+ properties. Maps Baxter material master data to standardized semantic categories — enabling cross-BL product comparison and external regulatory classification alignment.
VDI/VDE 3682	Process Ontology	German engineering standard for representing manufacturing processes as formal process models with inputs, outputs, and operators. Applied to Baxter pharmaceutical manufacturing processes: compounding, filling, lyophilization, sterilization, and packaging.
OPC-UA Companion Specifications	Equipment Semantics	Pharma-relevant specs: PDA TR-106 (pharmaceutical equipment capability representation), ISA-88 (equipment phases and procedural elements). Maps manufacturing equipment capabilities to ontological properties — enabling model-based process design and technology transfer.
OSDU Materials Ontology	Supply Chain Semantics	Adapted from the Open Subsurface Data Universe for pharma raw materials. Provides semantic classification of excipients, APIs, packaging materials, and intermediates — harmonizing Baxter's material master across BLs.
BOM Harmonization Engine	Cross-system Integration	XSLT and SWRL rule set that harmonizes Engineering BOM (Teamcenter), Manufacturing BOM (SAP), and Regulatory BOM (Veeva)

Component	Category	Description and Baxter Application
		into a unified product structure AAS. Resolves cross-system part number and version discrepancies through ontological equivalence mapping.

3.3.3 Baxter-Specific Considerations

The AAS layer addresses Baxter's fundamental multi-site, multi-BL challenge: identical equipment types run different PLC firmware at different sites, with different MES configurations and different historian tag naming conventions. The AAS provides site-agnostic semantic normalization — a Bioreactor at Site A is ontologically equivalent to a Bioreactor at Site B, regardless of vendor or configuration. This "semantic normalization without system harmonization" eliminates the need for expensive ERP or MES consolidation projects, delivering the cross-site analytics capability that consolidation would provide at a fraction of the cost and risk.

3.3.4 Key Architecture Decisions

- AAS Registry deployed on-premises within Baxter's secure manufacturing network; shells synchronized to FABRIC cloud knowledge graph via secure API
- Siemens AAS SDK (Java/Python) used for shell lifecycle management — avoids custom AAS tooling proliferation
- Baxter-specific submodels co-developed with Siemens Engineering Services, accelerating template delivery by 3-4 months
- ECLASS license included in Siemens IOX enterprise agreement — confirm coverage with Siemens contract team
- AAS shell creation automated for high-volume entities (materials, batches) via template instantiation; manual for equipment and facilities

Layer 4: Data Fabric

3.4.1 Purpose & Scope

The data fabric layer provides the physical storage and SQL transformation substrate for FABRIC. It implements an open data lakehouse architecture using Apache Iceberg as the universal table format, dbt Core for governed SQL transformations, and Trino/DuckDB as the federated query engine. This layer is responsible for data quality enforcement (Great Expectations), automated lineage capture (Apache Atlas), and the governed metric definitions (dbt Semantic Layer) that ensure all downstream analytics derive from consistent, version-controlled business logic. The data fabric is architecture-agnostic — it operates on S3, Azure ADLS, or on-premises MinIO without code changes, avoiding cloud vendor lock-in.

3.4.2 Component Inventory

Component	Category	Description and Baxter Application
Apache Iceberg	Table Format	ACID-compliant open table format with schema evolution, time-travel, and partition evolution. All FABRIC datasets stored as Iceberg tables on object storage. Time-travel is critical for pharmaceutical datasets where historical comparison (stability timepoints, annual product reviews) is a routine analytical need.
Delta Lake	Streaming Landing Zone	For Kafka-sourced streaming tables (Historian, real-time MES events). Delta Lake streaming ingest with ACID guarantees used as a landing zone before promotion to Iceberg governed tables. Decouples streaming ingestion latency from governance overhead.
dbt Core	SQL Transformation	Open-source SQL transformation framework. All FABRIC data transformations defined as dbt models with automated documentation, column-level lineage, and built-in testing. dbt Semantic Layer defines all canonical business metrics (batch yield, specification compliance rate, supplier lead time) as governed, version-controlled metric contracts.
Apache Spark / Trino	Query Engine	Spark for large-scale batch transformation jobs and ML feature engineering. Trino for interactive federated SQL across Iceberg tables spanning multiple storage locations — enables cross-BL analytical queries without physical data movement.
DuckDB	Embedded Analytics	In-process SQL engine for edge analytics. Enables manufacturing site-level analytics without cloud dependency. Governed Iceberg snapshots synchronized to local DuckDB instances — site engineers can run governed analytics during network outages.
Great Expectations	Data Quality	Data quality validation framework with 200+ built-in expectation types. FABRIC defines expectation suites per domain (R&D, manufacturing, supply chain, regulatory). Quality scores surface in OpenMetadata and trigger automated alerts on degradation. Expectation failures block data promotion to knowledge graph.
Apache Atlas	Lineage & Metadata	Automated data lineage capture from NiFi flow graphs through dbt transformation models to production dashboards. Provides full provenance: from SAP material master record through NiFi extraction, dbt transformation, Iceberg table, SPARQL query, to analyst dashboard cell.
OpenMetadata	Data Catalog	Unified metadata catalog with business glossary integration. Connects to Atlas lineage, dbt model documentation, and SHACL ontological definitions via API integration. Provides a single searchable surface for all Baxter data assets across all business lines.

3.4.3 Baxter-Specific Considerations

Baxter's multi-BL environment creates significant schema heterogeneity even within the same system category — two LIMS instances configured for different BLs have divergent table structures, terminology, and specification formats. The dbt Semantic Layer provides a governed abstraction over this heterogeneity: canonical metrics like yield, cycle time, and specification compliance rate are computed consistently regardless of source BL, using ontology-mapped business logic. This is the functional equivalent of Palantir's Metric Layer, implemented entirely on open-source tooling.

3.4.4 Key Architecture Decisions

- Apache Iceberg preferred over Delta Lake as primary format — avoids Databricks ecosystem dependency; runs on any compute engine
- dbt Core (self-hosted) preferred over dbt Cloud — eliminates per-seat recurring license cost; CI/CD via standard Git workflows
- Three storage tiers: hot (90 days, Trino-optimized), warm (3 years, Iceberg partitioned), cold (regulatory retention, object archive)
- All dbt models must have tests (`not_null`, `unique`, `accepted_values`) before promotion to production — enforced by CI gate
- Cross-BL joins prohibited in source-layer dbt models — cross-domain composition only at semantic layer above knowledge graph

Layer 5: Knowledge Graph

3.5.1 Purpose & Scope

The knowledge graph layer is the semantic intelligence core of FABRIC and the direct architectural analog of Palantir's Ontology layer — implemented using open W3C standards (RDF 1.1, OWL 2, SHACL, SPARQL 1.1, SWRL). It stores all Baxter entities as typed nodes with semantically defined, validated, and reasoned relationships, enabling cross-domain queries that no individual source system can answer. The knowledge graph is where the semantic harmonization layer's AAS entity models meet the data fabric's governed datasets — AAS shells become graph nodes, AAS submodel attributes become typed triples, and SHACL shapes enforce the integrity rules that GxP compliance requires. The graph is not a static repository but a living, reasoned, temporal model of Baxter's operational reality.

3.5.2 Component Inventory

Component	Category	Description and Baxter Application
Stardog Enterprise / GraphDB EE	Triplestore	Production-grade RDF graph database with integrated OWL 2 RL/EL reasoner. Stardog preferred for its native GraphQL interface, integrated ML inference, and enterprise security model. GraphDB Enterprise as fallback. Both support sub-second SPARQL on 10B+ triple graphs.
OWL 2 Domain Ontology	Formal Ontology	Baxter FABRIC upper ontology extending BFO (Basic Formal Ontology) with IDTA AAS-derived classes. Core pharma domain classes: PharmaceuticalProduct, ManufacturingBatch, ActivePharmaceuticalIngredient, ExcipientMaterial, ManufacturingSite, SupplierOrganization, RegulatoryDossier, ClinicalSpecification.
SHACL Constraint Library	Data Validation	Shapes Constraint Language rules enforcing data integrity across the graph. 200+ SHACL shapes covering: product-batch relationships, specification version constraints, supplier qualification temporal rules, batch genealogy completeness, and regulatory linkage requirements. SHACL violations block graph promotion.
SWRL Temporal Rules	Automated Inference	Semantic Web Rule Language rules for temporal reasoning — specification expiration, supplier qualification windows, stability interval enforcement, and retest date calculation. Enables automated flags: "specifications approaching expiry," "supplier qualifications due for renewal," "stability timepoints overdue."
SPARQL 1.1 Federated Endpoint	Query Interface	Federated SPARQL 1.1 endpoint accessible to analytics tools, LLM agents, and API consumers. SERVICE federation enables cross-triplestore queries — a single SPARQL query can span R&D, manufacturing, and supply chain named graphs. Named graph per source system for provenance tracking.
R2RML / RML Mapping	Data Loading	RDF Mapping Language specification transforming Iceberg tabular data into typed RDF triples. All mappings version-controlled in Git — testable, reviewable, and deployable via CI/CD. Enables incremental graph updates on each dbt model refresh cycle.
FAISS Semantic Search	Vector Search	Facebook AI Similarity Search index over entity embeddings (generated by SBERT on entity labels and descriptions). Enables semantic search ("find materials similar to Excipient X based on function") alongside precise SPARQL queries. Primary retrieval mechanism for LLM Gateway RAG augmentation.
VocBench 3 Ontology Registry	Governance Tool	Collaborative ontology authoring and governance platform. All ontology changes follow a peer-review workflow (propose → review → approve)

Component	Category	Description and Baxter Application
		→ deploy) with audit trail. Git integration for version control. Integration with SHACL validator for pre-deployment constraint testing.

3.5.3 Baxter-Specific Considerations

The knowledge graph is what gives FABRIC its "Palantir feel" for business users. A Baxter R&D scientist can ask: "Which batches of Product A failed dissolution testing, and what raw material lots did they share with passing batches?" — a query requiring manual cross-referencing across LIMS, MES, and SAP without FABRIC. With the knowledge graph, this is a single SPARQL query returning a traceable result set in seconds. The SHACL constraint library additionally serves as the formal, machine-executable specification of GxP data integrity rules — making the graph itself an auditable compliance artifact acceptable to FDA and EMA inspectors.

3.5.4 Key Architecture Decisions

- Named graphs per source system domain — provenance, timestamp, and version tagged on all triples at ingest
- OWL 2 RL profile for production reasoning (decidable, polynomial complexity); OWL 2 Full available for ontology development
- No destructive ontology updates — deprecated classes retained with owl:deprecated annotation and tombstone timestamps
- SPARQL queries against production graph limited to read-only SELECT and ASK — SPARQL UPDATE only on curated named graphs via governed workflow
- Knowledge graph version snapshots taken weekly to Iceberg — enables time-travel analytics at the graph level

Layer 6: AI & Analytics

3.6.1 Purpose & Scope

The AI and analytics layer delivers the intelligence surfaces consumed by data scientists, analysts, and AI-powered applications. It includes open-source business intelligence (Apache Superset), a full ML lifecycle platform (MLflow), a self-hosted generative AI gateway (vLLM/Ollama), AI agent orchestration (LangGraph), and production model serving (Ray Serve). All AI and ML models deployed in this layer are governed by the MTBH (Mean Time Between Hallucinations) Circuit Breaker framework, which provides continuous fidelity monitoring and automatic model suspension when confidence thresholds are violated. This governance architecture positions Baxter ahead of both FDA AI/ML guidance for pharmaceutical manufacturing and EU AI Act requirements for high-risk AI in healthcare.

3.6.2 Component Inventory

Component	Category	Description and Baxter Application
Apache Superset	Business Intelligence	Open-source BI platform with 40+ chart types, SQL Lab for ad-hoc exploration, and native dbt Semantic Layer integration. All production dashboards connect to governed dbt metrics — no ad-hoc SQL metric definitions. Role-based access control aligned to Baxter organizational structure and ontological role definitions.
MLflow	ML Lifecycle Management	End-to-end ML experiment tracking, model registry, and deployment management. Every training run logged with hyperparameters, code version, data version, and evaluation metrics. Model promotion workflow: experiment → candidate → staging (validation) → production (MTBH-monitored). Replaces all shadow ML tracking systems.
JupyterHub	Data Science Workbench	Multi-user Jupyter environment with pre-configured kernels (SPARQL, pandas, scikit-learn, PyTorch). FABRIC Iceberg datasets auto-mounted via PyIceberg connector. Knowledge graph accessible via SPARQLWrapper. Replaces Palantir Code Workbook at zero license cost.
LLM Gateway (vLLM / Ollama)	Generative AI Infrastructure	Self-hosted LLM inference server supporting multiple models simultaneously. Selected models: Llama 3.1 70B (general knowledge work), BioMedLM (biomedical text understanding), and Baxter-specific fine-tunes for regulatory/CMC text. All inference on Baxter-controlled infrastructure — no PII or trade secrets sent to external API providers.
LangGraph Agents	AI Agent Orchestration	Multi-step AI agent framework for complex knowledge tasks. Primary agents: Batch Investigation Agent (traverses batch genealogy, LIMS, and process data), Regulatory Evidence Agent (retrieves and cross-references CMC documentation), Supply Chain Risk Agent (multi-tier supplier risk assessment). All agents operate over SPARQL + Iceberg + document retrieval.
FLAML / AutoML	Automated Machine Learning	Fast and lightweight AutoML library. Enables R&D scientists without ML expertise to train predictive models on governed Baxter datasets. Primary use cases: batch yield prediction, stability outcome forecasting, equipment failure prediction, and specification acceptance rate modeling.
Ray Serve	Production Model Serving	Distributed model serving framework with autoscaling, model versioning, and A/B traffic splitting. MTBH telemetry hooks integrated at the serving layer — every inference recorded for fidelity monitoring. Supports both ML models (from MLflow) and LLM chains (from LangGraph).

Component	Category	Description and Baxter Application
Grafana	Operational Monitoring	Real-time operational dashboards for manufacturing KPIs, process parameter monitoring, equipment OEE, and FABRIC platform health. Connects to Delta Lake time-series tables and Historian data via ClickHouse adapter. Separate from Superset — Grafana serves operations; Superset serves analysis.

3.6.3 Baxter-Specific Considerations

For Baxter R&D, the highest-value AI use cases in Phase 1 are: (1) Formulation Knowledge Synthesis — LLM agent querying the knowledge graph to surface relevant prior formulation work and failed experiments across all BLs for a given new program; (2) Batch Investigation Assistant — multi-step agent performing root cause analysis by traversing batch genealogy, LIMS results, and process historian data simultaneously; (3) Stability Trend Forecasting — FLAML models predicting out-of-specification risk early in stability studies using historical stability outcome patterns. The MTBH Circuit Breaker ensures none of these AI outputs reach users below the configured confidence threshold for each use case.

3.6.4 Key Architecture Decisions

- Self-hosted LLMs only for all production use cases — no external LLM API calls with Baxter data in any phase
- RAG architecture over knowledge graph preferred over fine-tuning for regulated use cases — preserves answer traceability to source documents
- MTBH threshold calibrated per use case risk class: patient-safety-adjacent use cases require higher MTBH than operational optimization
- All Superset dashboards connected to dbt Semantic Layer — ad-hoc dashboard SQL flagged as "uncertified" for user awareness
- LangGraph agent actions logged to immutable audit log for all regulated workflows — required for 21 CFR Part 11 compliance

Layer 7: Capability Applications

3.7.1 Purpose & Scope

The capability applications layer represents the business-facing intelligence surfaces of FABRIC — the dashboards, decision support tools, process automation workflows, and self-service analytics that deliver measurable business outcomes to Baxter employees across R&D, manufacturing, supply chain, quality, and regulatory functions. These applications compose capabilities from the lower six layers into purpose-built user experiences. The design principle is "configure, don't code" — all capability applications assemble from FABRIC platform components rather than building net-new technology. Each application is treated as an independent product with its own Product Owner, user research cycle, adoption metric, and iteration cadence.

3.7.2 Component Inventory

Component	Category	Description and Baxter Application
R&D Intelligence Platform	R&D	Cross-BL research knowledge surface. Natural language query over formulation history, stability databases, and analytical method repositories across all business lines. Patent landscape visualization. Prior art synthesis for new program initiation. Failed experiment knowledge preservation and cross-program learning.
Batch Intelligence & QC Release Support	Manufacturing/Quality	Real-time batch status tracking across all facilities. Automated QC release decision support with SHACL-validated linkage of LIMS analytical results to MES batch records. 21 CFR Part 11 compliant electronic review and signature workflow. Batch genealogy visualization from API to finished product.
Supply Chain Risk Observatory	Supply Chain	Multi-tier supplier visibility dashboard. Real-time supplier qualification status, lead time intelligence, geopolitical risk scoring, and specification compliance monitoring across all BLs. Concentration risk analysis (single-source API suppliers). CMO performance trending and capacity modeling.
Manufacturing Process Analytics	Manufacturing	OEE, cycle time, yield, and Continued Process Verification (CPV) dashboards. Equipment performance trending with predictive maintenance indicators. Process deviation detection with automatic knowledge graph cross-reference to prior deviations and their resolutions.
Regulatory Intelligence Hub	Regulatory/CMC	CMC data traceability from development batch to regulatory submission. Specification version history with full audit trail. Regulatory change monitoring (FDA guidance, EMA updates, ICH harmonization) with impact analysis against Baxter's product portfolio. eCTD section population assistance via LangGraph Regulatory Evidence Agent.
Quality & Compliance Command Center	Quality	Integrated CAPA status, deviation trending, and change control pipeline visualization. Audit readiness scoring by site and product line. Cross-BL quality KPI benchmarking. Predictive quality risk model for proactive deviation prevention. Inspection readiness package generation.
Process Automation Workflows	Operations	LangGraph-powered workflow automation: regulatory template population, batch record review assistance, supplier audit scheduling, stability report generation, and specification comparison. Human-in-the-loop gates mandatory for all regulated decision outputs — no autonomous action on patient-safety decisions.

Component	Category	Description and Baxter Application
Self-Service Analytics Portal	All Users	Governed self-service analytics for business users without data science backgrounds. Pre-certified data domains, curated starting dashboards, and a guided query builder that generates SPARQL/SQL from natural language via LLM agent. All outputs watermarked "Self-Service / Unvalidated" until certified.

3.7.3 Baxter-Specific Considerations

Capability applications are the primary mechanism by which Baxter business users experience FABRIC — and therefore the primary driver of adoption and realized business value. Application delivery is sequenced by business value and technical dependency: R&D Intelligence and Batch Intelligence in Phase 1 (highest immediate value, cleanest data), Supply Chain Observatory and Manufacturing Analytics in Phase 2 (broader integration needed), Regulatory Intelligence and Quality Command Center in Phase 3 (highest complexity, regulatory sensitivity). Each application has a designated Baxter Product Owner who owns the user adoption metric and is accountable to the Business Change Network.

3.7.4 Key Architecture Decisions

- Each capability application has a dedicated Product Owner from the relevant Baxter business line — architecture team is not the product owner
- MVP shipped at end of each phase — no waterfall delivery within phases; iterate bi-weekly on user feedback after MVP
- Adoption metric: 60% weekly active usage rate by month 3 post-launch — applications below threshold trigger a redesign sprint
- Process automation limited to non-patient-safety decisions in Phase 1; Phase 3 allows AI-assisted (HITL) regulated outputs
- All capability applications use FABRIC authentication/authorization — no application-level user management

4. Four-Domain Governance Framework

The FABRIC governance framework spans all seven architecture layers and provides the institutional structures, tooling, and enforcement mechanisms required for a compliant, trustworthy enterprise intelligence platform in a regulated pharmaceutical and medical device environment. The four domains are designed to interlock — semantic governance defines the concepts that data governance catalogs, analytics governance builds on the data that data governance validates, and AI governance monitors the models that analytics governance promotes.

Domain 1: Data Governance

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OpenMetadata · Apache Atlas · Great Expectations · OPA Policy Engine · SAP MDM

Domain Mandate

Establish and maintain data as a trusted, catalogued, quality-scored, and lineage-tracked asset across all Baxter systems and business lines. Data governance operationalizes the principle that data quality is a prerequisite for knowledge quality — no data enters the knowledge graph without passing its Great Expectations validation suite.

Core Capabilities

- Unified data catalog spanning all source domains with automated discovery and schema profiling
- End-to-end automated data lineage: from SAP database table through NiFi extraction, dbt transformation, Iceberg storage, to Superset dashboard cell
- Data Quality Score (DQS) calculated per dataset, per business line, per refresh window — exposed in OpenMetadata and dashboard watermarks
- Master data management (MDM) for product, supplier, facility, and organizational master data — single authoritative record across all BLs
- PII and sensitive data classification with OPA policy enforcement — attribute-based access control aligned to ontological role definitions
- Data retention, archiving, and purge automation compliant with 21 CFR Part 11, EU GDPR, and Baxter corporate data policy
- Data stewardship program: designated stewards per domain with defined accountability metrics and escalation paths

Governance Metrics

- Catalog coverage: percentage of active production datasets catalogued (target: 95% by end of Phase 2)
- Data Quality Score: composite DQS per critical dataset (target: >92% for R&D datasets by Phase 1 completion)
- Lineage completeness: percentage of production reports with end-to-end lineage documented (target: 100% by Phase 3)

Domain 2: Semantic Governance

TOOLING

VocBench 3 · Protégé · OntoDB · SHACL Validator · Git (ontology repository)

Domain Mandate

Maintain a coherent, versioned, organizationally adopted ontological model that accurately represents Baxter's business concepts, processes, products, and relationships. Semantic governance ensures that the knowledge graph remains a shared understanding, not a Tower of Babel, as business requirements evolve and new domains are added.

Core Capabilities

- Centralized ontology registry with Git-backed versioning and full change history — every class, property, and axiom tracked
- Controlled vocabulary management per domain (product, process, organizational, regulatory) with alignment to external standards
- Formal term lifecycle: proposed → under ontology committee review → approved → stable → deprecated (with sunset timeline)
- SHACL constraint library (200+ shapes) enforcing cross-domain data integrity invariants — pre-deployment testing mandatory
- External ontology alignment: IDTA/ECLASS (manufacturing), SNOMED CT (clinical), NCI Thesaurus (regulatory), MeSH (biomedical)
- Ontology change impact analysis: automated SPARQL query identifying all downstream artifacts affected by proposed changes
- Ontology Governance Committee: cross-functional bi-weekly review body with Architecture Board tie-breaking authority

Governance Metrics

- Ontology coverage: percentage of critical business concepts formally defined in the domain ontology (target: 80% by Phase 2)
- SHACL compliance rate: percentage of knowledge graph triples passing all applicable SHACL constraints (target: >99.5% by Phase 3)
- Ontology adoption: percentage of capability applications consuming governed ontology terms vs. local concept copies (target: 100%)

Domain 3: Analytics Governance

TO

MLflow Model Registry · dbt Semantic Layer · Apache Superset RBAC · Great Expectations · Audit Logs

Domain Mandate

Ensure all analytics, metrics, and ML models consumed by Baxter business users are accurate, traceable, consistent, and appropriately controlled. Analytics governance eliminates metric drift — the common failure mode where the same business metric (e.g., "batch yield") is calculated differently in different dashboards, eroding trust and driving manual reconciliation work.

Core Capabilities

- ML experiment registry: every training run tracked with full hyperparameter, code version, training data version, and evaluation metric lineage
- Model promotion workflow: experiment → candidate → staging (statistical validation) → production (MTBH-monitored, auto-rollback capable)
- dbt Semantic Layer as single source of truth for all business metric definitions — metric drift structurally impossible for certified dashboards
- Dashboard certification program: "Baxter Certified" badge on dashboards passing data quality, business logic, and peer review gates
- Role-based access control via semantic policies: ontological role assignments drive data access — no manual permission matrix management
- Statistical validation gates: A/B testing framework for model comparison before any production promotion
- Analytics audit log: all dashboard views, report exports, and model inferences logged for regulatory readiness and usage analytics

Governance Metrics

- Metric consistency: percentage of production dashboards sourcing metrics from dbt Semantic Layer (target: 100% for certified dashboards)
- Model promotion cycle time: days from model training completion to production deployment (target: under 10 business days by Phase 2)
- Certification rate: percentage of high-use dashboards achieving Baxter Certified status (target: 85% by end of Phase 3)

Domain 4: AI Governance (MTBH Framework)

MTBH Circuit Breaker · OPA Policy Engine · Model Cards · EU AI Act Readiness Scanner · Immutable Audit Log

Domain Mandate

Ensure all AI systems deployed within FABRIC are trustworthy, fidelity-monitored, traceable, and compliant with emerging FDA and EU AI regulatory requirements. The MTBH (Mean Time Between Hallucinations) metric provides a formal, quantitative trust measure for all generative AI and ML model deployments — operationalizing AI trustworthiness as an engineering discipline rather than a qualitative aspiration. The Circuit Breaker architecture enforces automatic model suspension before unreliable outputs reach Baxter users.

Core Capabilities

- MTBH metric calculation per deployed model per deployment window — updated continuously, surfaced in model registry and capability app health dashboards
- Circuit Breaker architecture: automated model suspension triggered when MTBH falls below use-case-specific threshold — zero-downtime failover to last stable model version
- Composition Theorem enforcement: multiplicative fidelity degradation tracked across multi-model pipelines — prevents compounding of individual model uncertainty
- Sanitization framework: input and output validation for all LLM inferences — prevents semantic drift and prompt injection in production workflows
- Model cards: bias assessments, intended use declarations, known limitations, and out-of-scope warnings for all production models — published to model registry
- Human-in-the-loop (HITL) gates mandatory for all AI outputs with patient safety implications — no autonomous action on regulated decisions
- EU AI Act Article 9 risk management: risk classification per AI system, technical documentation, post-market monitoring plan for applicable systems
- FDA AI/ML SaMD readiness assessment for Baxter AI applications meeting Software as Medical Device criteria — predetermined change control protocol

Governance Metrics

- MTBH per model: mean time between confidence threshold violations — calibrated per use case risk class (patient safety vs. operational)
- Circuit Breaker interventions: number of automated model suspensions per month — leading indicator of model health degradation
- HITL compliance: percentage of regulated AI decisions receiving documented human review (target: 100% with zero exceptions)
- EU AI Act readiness score: composite compliance score across all applicable high-risk AI systems (target: >85% by Phase 3)

5. Implementation Roadmap

The FABRIC implementation is organized into three phases over 36 months, each building on the preceding phase's infrastructure and organizational capability. The phased approach enables Baxter to demonstrate measurable value within 12 weeks (R&D intelligence) while building the institutional infrastructure required for enterprise-scale deployment. Each phase concludes with a formal Architecture Gate Review — a structured go/no-go decision requiring evidence of technical delivery, user adoption, and governance readiness before the next phase is funded.

Phase	Timeline	Focus	Primary Deliverables	Indicative Investment
Phase 1: Foundation	Months 1–6	R&D intelligence, core infrastructure, governance baseline	Layers 1–5 operational for R&D; 2 capability apps live; governance baseline	\$600K – \$900K
Phase 2: Scale	Months 7–18	Manufacturing, supply chain expansion, portfolio rationalization Wave 1-2	All 7 layers operational for all BLs; 6 capability apps; Portfolio Wave 1-2 complete	\$800K – \$1.2M
Phase 3: Optimize	Months 19–36	AI augmentation, regulatory intelligence, knowledge federation, full governance maturity	All 8 apps; full governance maturity; EU AI Act compliance; legacy decommission complete	\$500K – \$800K/yr

5.1 Phase 1: Foundation (Months 1–6)

Phase 1 establishes the architectural foundation and delivers the first two capability applications to demonstrate business value within 12 weeks. Success is defined by delivering R&D Intelligence and Batch Intelligence to 20+ active users with measurable time savings in cross-BL formulation research and batch QC release. Every Phase 1 activity is sequenced to build reusable infrastructure — AAS models, ontology foundations, and data pipelines established in Phase 1 extend directly into Phase 2 without rework.

Month(s)	Workstream	Activity	Deliverable	Owner
1	Infrastructure	Cloud environment provisioning, network security zones, CI/CD pipeline, GitOps repository structure	FABRIC infrastructure baseline — all teams can deploy	Platform Architect
1–2	Layer 1–2	SAP S/4HANA and LIMS integration: NiFi flows, Airbyte connectors, Kafka topics for R&D domains	R&D source data flowing into staging zone	Integration Lead
1–2	Layer 3 (Siemens)	Siemens AAS Registry deployment; IDTA submodel instantiation for R&D assets; IOX configuration for R&D facility	AAS shells for materials, products, equipment (R&D scope)	Semantic Architect + Siemens
2–3	Layer 4	Iceberg table design for R&D domain; dbt models for ELN, LIMS, SAP R&D objects; Great Expectations suites	Governed R&D data domain in Iceberg — data quality >90%	Data Engineer
3–4	Layer 5	Seed ontology from IDTA/ECLASS; initial OWL 2 class hierarchy for R&D	R&D Knowledge Graph v1.0 — queryable by R&D users	Knowledge Engineer

Month(s)	Workstream	Activity	Deliverable	Owner
		domain; 50+ SHACL shapes; SPARQL endpoint		
3–4	Layer 6	MLflow deployment; JupyterHub with FABRIC kernels; Superset connected to dbt Semantic Layer; MTBH baseline	Analytics workbench operational for R&D scientists	Analytics Lead
4–5	Layer 7	R&D Intelligence MVP: cross-BL formulation query interface, ELN harmonization view, failed experiment search	R&D Intelligence App v1.0 — live for R&D pilot users	Product Owner: R&D
5–6	Layer 7	Batch Intelligence MVP: LIMS-MES semantic linkage, batch genealogy graph, QC release decision support	Batch Intelligence App v1.0 — live for QC/Manufacturing pilot	Product Owner: Manufacturing
5–6	Governance	OpenMetadata catalog seeding; initial SHACL constraints deployed; MTBH baselines set; Ontology Committee chartered	Governance baseline — all four domains operational at Level 1	Data Governance Lead
6	PMO	Phase 1 Gate Review: adoption evidence, technical validation, governance readiness, Phase 2 business case sign-off	Phase 1 Go/No-Go decision — Phase 2 funding authorization	Program Director

GATE 1

Phase 1 → Phase 2 Criteria: (1) 20+ weekly active users on R&D Intelligence; (2) LIMS-MES batch graph operational and validated; (3) Governance baseline approved by Data Governance Council; (4) Data Quality Score >90% for R&D domain; (5) Phase 2 funding authorized by Steering Committee.

5.2 Phase 2: Scale (Months 7–18)

Phase 2 extends FABRIC from R&D to manufacturing, supply chain, and quality domains. The Siemens AAS semantic layer expands from the R&D pilot facility to all manufacturing sites. Supply chain digital twins are created for Tier 1 and Tier 2 suppliers. The knowledge graph grows from approximately 10M to 500M+ triples. Portfolio rationalization Wave 1 and Wave 2 begin decommissioning duplicate analytical tools and shadow data warehouses. By Month 18, all seven layers are operational across all Baxter business lines.

Quarter	Workstream	Activity	Deliverable	Owner
Q3 (M7–9)	Layer 1–2 Manufacturing	Opcenter MES, PI Historian, SCADA integration via Siemens IOX; manufacturing NiFi flows	Manufacturing data domain flowing into FABRIC	Integration Lead
Q3 (M7–9)	Layer 1–2 Supply Chain	SAP APO/IBP integration; CMO and supplier portal data feeds via Kong API Gateway	Supply chain data domain flowing into FABRIC	Integration Lead
Q3–Q4 (M8–12)	Layer 3 (Siemens) Expand	AAS shells for all manufacturing assets across all sites; IOX operational data context; Opcenter MES AAS linkage	Full-facility digital twin coverage — all manufacturing sites	Semantic Architect + Siemens

Quarter	Workstream	Activity	Deliverable	Owner
Q4 (M10–12)	Layer 5 Expand	Manufacturing ontology extension (batch, equipment, process); supply chain ontology (supplier, material, logistics)	Knowledge graph extended to manufacturing and supply chain	Knowledge Engineer
Q4–Q5 (M10–15)	Layer 7 — Apps 3 & 4	Manufacturing Process Analytics MVP; Supply Chain Risk Observatory MVP	3rd and 4th capability applications live	Product Owners
Q5 (M13–15)	Governance Level 2	Full lineage graph deployed; RBAC rollout to all domains; Circuit Breaker on all production AI models; SOPs for all governance bodies	Phase 2 governance milestone — all four domains at Level 2	Data Governance Lead
Q5–Q6 (M14–18)	Layer 7 — Apps 5 & 6	Quality & Compliance Command Center MVP; Process Automation Workflows MVP	5th and 6th capability applications live	Product Owners
Q6 (M16–18)	Portfolio Rationalization	Standalone BI tools migrated to Superset; shadow data warehouses migrated to Iceberg; decommission Wave 1-2 complete	Wave 1-2 decommissions complete — \$450K–\$1M annual savings	PMO Director + IT
M18	PMO	Phase 2 Gate Review: adoption metrics, KPI performance, portfolio progress, Phase 3 roadmap approval	Phase 2 Go/No-Go — Phase 3 planning authorized	Program Director

GATE 2

Phase 2 → Phase 3 Criteria: (1) All 6 capability apps at >60% weekly active adoption; (2) Supply chain digital twin covering 80%+ of Tier 1 suppliers; (3) Full data lineage documented for all production reports; (4) MTBH baselines established and Circuit Breaker active for all production AI models; (5) Portfolio Wave 1-2 decommissions confirmed.

5.3 Phase 3: Optimize (Months 19–36)

Phase 3 transitions FABRIC from deployment to optimization and strategic expansion. AI augmentation deepens across all capability applications. Regulatory intelligence use cases are activated. Cross-enterprise knowledge federation is extended to select CMO partners and key API suppliers via AAS-based federated integration. Full governance maturity is achieved across all four domains. Portfolio rationalization Wave 3 completes legacy decommissioning and targets SAP BW reduction.

Timeline	Theme	Key Activities	Strategic Outcome
M19–24	AI Augmentation	LangGraph agent deployment for batch investigation, formulation synthesis, regulatory template generation; MTBH threshold tuning per use case risk class; FLAML predictive models in production	AI-augmented R&D and manufacturing workflows with MTBH-guaranteed fidelity
M21–27	Regulatory Intelligence	SPL/eCTD ontology development; CMC traceability graph from development to submission; Veeva RIM integration; FDA/EMA regulatory change monitoring feeds; Regulatory Evidence Agent deployment	Automated CMC data traceability; eCTD section population assistance; regulatory change impact analysis

Timeline	Theme	Key Activities	Strategic Outcome
M24–30	Knowledge Federation	AAS-based federated integration with select CMO partners and Tier 1 API suppliers; SPARQL federation endpoint for cross-enterprise query; partner data quality SLA agreements	Ecosystem-level supply chain intelligence; CMO process visibility without direct system access
M27–33	Governance Maturity	Self-service ontology extension toolkit for power users; predictive data quality monitoring; EU AI Act Article 9 compliance documentation for applicable systems; FDA AI/ML predetermined change control protocols	Full governance maturity (Level 3 all four domains); regulatory AI compliance posture established
M30–36	Optimization & Rationalization	Legacy decommission Wave 3 (reporting databases, SAP BW reduction); infrastructure cost optimization (right-sizing, spot instances); advanced CPV analytics; cross-BL benchmarking suite	Full TCO target achieved; \$720K–\$1.7M annualized rationalization savings realized

6. Portfolio Rationalization

FABRIC does not replace Baxter's systems of record — it creates a semantic intelligence layer above them. However, FABRIC's deployment creates the opportunity to rationalize the significant number of duplicate analytical tools, shadow data warehouses, ad-hoc reporting databases, and disconnected ML experimentation platforms that have accumulated across Baxter's business lines. Portfolio rationalization is sequenced in three waves to capture license savings without disrupting operational systems.

6.1 Rationalization Principles

- Systems of Record (ERP, MES, LIMS, QMS, PLM) are preserved — FABRIC reads from them and never replaces them in Phase 1 or Phase 2
- Duplicate analytical tools, reporting databases, and shadow data marts are the primary rationalization targets
- GxP-validated systems require formal validation impact assessment and change control before any modification — rationalization decisions involving these systems are deferred to Phase 3
- Usage analytics inform rationalization priority — systems with <20% active usage at time of FABRIC equivalent delivery are first wave candidates
- Rationalization decisions are issued as formal Architecture Decision Records (ADRs) approved by the Architecture Board
- Migration validation precedes decommission — parallel operation period (minimum 90 days) required before any production system is turned off

6.2 System Rationalization Decisions

System / Tool	Layer Overlap	Decision	Rationale	Wave	Annual Savings Est.
Standalone BI tools (Tableau, Power BI scatter licenses)	L6–L7	CONSOLIDATE to Superset	FABRIC Superset + dbt Semantic Layer provides governed equivalent; eliminate per-seat license proliferation across BLs	Wave 2	\$200–400K
Shadow data warehouses (BL-specific Redshift, Snowflake instances)	L4	MIGRATE to Iceberg, DECOMMISSION	Replaced by FABRIC data fabric on Apache Iceberg; maintain read replicas during 90-day migration validation period	Wave 2–3	\$150–350K
Disconnected ELN analytics (Excel, custom Python scripts, Access)	L5–L7	REPLACE by R&D Intelligence App	R&D Intelligence Platform provides governed, cross-BL equivalent — eliminates shadow analytics and associated data integrity risk	Wave 1	\$50–100K (labor)
Duplicate ontology tooling (local OWL editors, ad-hoc taxonomies)	L3–L5	CONSOLIDATE to VocBench + OntoDB	Centralized ontology governance under FABRIC semantic governance domain; local copies prohibited after transition	Wave 1	\$20–60K
Legacy reporting databases (Oracle SSRS, Access, Crystal Reports)	L4–L7	MIGRATE to dbt + Superset, DECOMMISSION	All operational reports migrated to dbt models on Iceberg; legacy reporting infrastructure eliminated by Phase 3	Wave 2–3	\$100–200K

System / Tool	Layer Overlap	Decision	Rationale	Wave	Annual Savings Est.
Scattered ML platforms (Databricks, notebooks, local MLflow instances)	L6	CONSOLIDATE to FABRIC MLflow	Single ML registry reduces governance complexity; eliminates shadow model proliferation; centralized MTBH monitoring	Wave 2	\$100–300K
SAP S/4HANA ERP	L1	RETAIN (System of Record)	Core ERP not displaced. Evaluate SAP BW (Business Warehouse) decommission in Wave 3 when Iceberg analytics coverage is confirmed sufficient	Long-term	BW savings TBD
Siemens Opcenter MES	L1–L2	RETAIN + UPGRADE with IOX	Core MES retained; Siemens IOX upgrade unlocks semantic integration. Validates enterprise agreement; may qualify for bundled pricing	Phase 1	Bundled savings
Siemens Teamcenter PLM	L1–L2	RETAIN (System of Record)	PLM not displaced; AAS submodels read from Teamcenter BOM and DHF. Evaluate Teamcenter Active Workspace redundancy in Phase 2	Phase 2 eval	\$50–150K
Veeva Vault QMS / RIM	L1	RETAIN (System of Record)	Regulated document management not displaced; FABRIC provides traceability, analytics, and regulatory intelligence on top of Veeva data	Long-term	N/A

6.3 Rationalization Value Summary

Wave	Timeline	Primary Targets	Estimated Annual Savings
Wave 1	Phase 1 (M4–6)	Disconnected ELN analytics, ad-hoc scripts, duplicate ontology tooling	\$70K – \$160K
Wave 2	Phase 2 (M10–18)	Standalone BI licenses, shadow data warehouses, duplicate ML platforms	\$450K – \$1.05M
Wave 3	Phase 3 (M24–36)	Legacy reporting databases, SAP BW (partial), remaining scattered tools	\$200K – \$500K
Total rationalization (by M36)	Cumulative	All three waves combined	\$720K – \$1.71M / year

TCO NOTE

Rationalization savings are net of FABRIC platform operational cost increases. The combined effect targets a TCO-neutral or better position by end of Phase 2 (versus Baxter's pre-FABRIC analytical tool spend), with progressive net cost reduction through Phase 3. The FABRIC + rationalization combined annual savings versus a comparable Palantir Foundry deployment is estimated at \$6–13M per year at steady state.

7. Enterprise PMO Structure

The Baxter FABRIC program requires a dedicated Enterprise PMO structure that bridges technology execution with business adoption and governance. The model follows a federated design — a centralized Program Management Office coordinates across four business-line workstreams and six technical workstreams, with clear escalation paths, decision rights, and accountability at each level. The PMO structure is designed to scale: Phase 1 operates lean, expanding in Phase 2 as scope broadens to manufacturing and supply chain.

7.1 Program Governance Model

Governance Body	Composition	Cadence	Decision Rights
Executive Steering Committee	CTO, CDO, CFO, VP R&D, VP Manufacturing, VP Supply Chain (Baxter); TVE Group Principal	Quarterly (+ ad-hoc for Phase Gates)	Phase gate approvals; budget authorization >\$500K; strategic direction; executive escalation resolution
FABRIC Architecture Board	Enterprise Architect (TVE); Solution Architect; Data Architect; Security Architect; Siemens Solutions Partner	Bi-weekly	All architecture decisions; technology selection; design exception approval; cross-workstream dependency resolution; ADR approval
Data Governance Council	Data Governance Lead; Data Stewards per BL; Compliance; Legal; IT Security; Ontology Committee Chairs	Bi-weekly	All data and ontology policy decisions; SHACL constraint library governance; ontology change approvals; access control policy
PMO Operations Board	Program Director; Workstream Leads (x6); Business Product Owners (x4)	Weekly	Sprint planning; issue and risk escalation; resource allocation; milestone tracking; stakeholder communications
Business Change Network	BL Champions (R&D, Manufacturing, SC, Quality); Super Users; Training Coordinators	Monthly	Adoption feedback; change readiness assessment; training needs; capability application prioritization; user community governance

7.2 Program Team Structure & Roles

Role	Type	Core Responsibilities	FTE	Source
Program Director	Leadership	Overall program accountability; executive stakeholder management; Phase gate ownership; budget oversight; Steering Committee interface	1.0	TVE Group / Baxter Senior Leader
Enterprise Architect (FABRIC)	Architecture	FABRIC architecture integrity; layer design decisions; cross-layer dependency management; ADR authoring; Siemens partner liaison	1.0	TVE Group
Solution Architect (Semantic)	Architecture	Knowledge graph design; OWL ontology development; AAS/IDTA integration design; SHACL library ownership; SPARQL performance	1.0	TVE Group / Semantic Specialist

Role	Type	Core Responsibilities	FTE	Source
Data Architect	Architecture	Iceberg/data fabric design; dbt modeling standards; lineage architecture; OpenMetadata configuration; data domain governance	1.0	TVE Group / Internal IT
Integration Lead	Technical	NiFi/Kafka/Airbyte pipeline development; source system integration patterns; IOX/OPC-UA configuration; connector library management	1.5	Internal IT / Systems Integrator
Data Engineers (x2)	Technical	dbt model development; Great Expectations suite authoring; Iceberg table management; pipeline operationalization	2.0	Internal IT / Systems Integrator
Knowledge Engineer	Technical	OWL ontology authoring; SHACL constraint development; SPARQL query optimization; R2RML mapping; VocBench administration	1.0	TVE Group / Specialist
ML / AI Engineer	Technical	MLflow configuration; model training pipelines; LLM gateway setup; MTBH instrumentation; Ray Serve deployment; agent development	1.0	Internal AI CoE / Specialist
Analytics Engineer	Technical	Superset dashboard development; dbt Semantic Layer metric definitions; JupyterHub administration; end-user training delivery	1.0	Internal IT
PMO Coordinator	PMO	Project tracking and reporting; meeting facilitation; risk register maintenance; budget tracking; communications management	1.0	Internal PMO
Data Governance Lead	Governance	OpenMetadata and Atlas administration; data stewardship program; DQS monitoring; governance policy development and enforcement	1.0	Internal / TVE Group
Security Architect (part-time)	Security	FABRIC security architecture review; OPA policy review; network segmentation validation; GxP audit readiness support	0.5	Internal IT Security
Siemens Solutions Partner	Partner	AAS registry deployment; IOX configuration; Opcenter MES integration; IDTA submodel template development and validation	Variable (T&M)	Siemens Engineering Services
Product Owners (x4)	Business	Capability application requirements; user story prioritization; UAT sign-off; adoption accountability — one per business line	4.0 (0.5 each)	Baxter Business Lines
Business Analysts (x2)	Business	Process mapping; requirements elicitation; user acceptance testing; training material development; change management support	2.0	Internal / Contracted

7.3 RACI Matrix — Key Program Decisions

Decision	Exec Sponsor	Prog Director	Ent Architect	Gov Council	BL Product Owners
Phase gate approval	A	R	C	C	I
Technology selection (layer component)	I	C	A/R	C	I
Ontology class / property change	I	I	C	A/R	C
Source system integration priority	C	A	R	I	C
Capability app MVP scope definition	I	C	C	I	A/R
System decommission decision	A	R	C	C	C
Budget reallocation (>\$100K)	A	R	I	I	I
AI model production promotion	I	C	C	A/R	I
MTBH threshold configuration	I	C	A/R	R	C
Regulatory evidence package approval	I	I	I	A/R	R
SHACL constraint deployment	I	I	A/R	R	C
Architecture Decision Record (ADR)	I	C	A/R	C	I

RACI Key: R = Responsible (does the work) | A = Accountable (final authority) | C = Consulted (input required) | I = Informed (notified of outcome)

7.4 Program Reporting Cadence

Meeting / Report	Frequency	Audience	Key Outputs
Program Status Report	Weekly	PMO Operations Board	Milestone RAG ratings; sprint velocity; top 5 risks; budget actuals vs. plan; action items
Architecture Review	Bi-weekly	FABRIC Architecture Board	Design decisions log; ADR updates; cross-layer dependency map; technical debt register; Siemens partner status
Governance Council Meeting	Bi-weekly	Data Governance Council	DQS dashboards; ontology change requests; policy decisions; SHACL compliance metrics; data steward reports
Executive Dashboard	Monthly	Steering Committee	Program health scorecard; business value realized vs. business case; adoption metrics per app; Phase gate forecast; budget status
Business Change Network	Monthly	BL Champions and Super Users	Adoption metrics; user feedback synthesis; training completion rates; upcoming release previews; change impact reviews

Meeting / Report	Frequency	Audience	Key Outputs
Quarterly Business Review	Quarterly	Full Steering Committee	TCO vs. plan; value realization vs. business case; roadmap adjustments; resourcing decisions; Phase gate status
Phase Gate Review	Per phase	Steering Committee + Architecture Board	Phase deliverable evidence package; Go/No-Go recommendation; next phase budget authorization; resource plan

8. Risk Register

The following risks represent the highest-priority program risks identified during architecture development. Each risk includes a likelihood/impact assessment and a specific mitigation strategy. The risk register is maintained as a living document in the PMO tracking system, reviewed at each weekly PMO Operations Board and formally updated monthly.

Risk	Category	L	I	Mitigation Strategy
Source system access restrictions delay integration timelines	Technical / IT	H	H	Establish formal data access governance agreements with Baxter IT in Month 1. Pre-negotiate read-only service accounts for all 10 priority source systems before kickoff. Escalation path to CTO if access blocked more than 10 business days.
Siemens AAS adoption complexity greater than modeled	Technical	M	H	Begin with AAS shells for 10-15 highest-value asset types only; defer full coverage to Phase 2. Engage Siemens Engineering Services for on-site AAS deployment support from Month 1. Siemens has pre-built pharma AAS templates that reduce development effort by 60%.
Ontology Governance Committee reaches stalemate on terminology	Organizational	M	M	Pre-define ontology decision protocol (lazy consensus with 5-day comment window). Architecture Board has explicit tie-breaking authority over ontology disputes. Time-box term debates to 2 bi-weekly meeting cycles maximum before escalation.
Source system data quality worse than expected	Data	H	M	Phase 1 includes formal data quality assessment before knowledge graph population. Great Expectations suites deployed as first Layer 4 activity. DQS <80% on any critical dataset is a go/no-go criterion for knowledge graph promotion — prevents "garbage in, garbage out" at the semantic layer.
GxP validation requirements for FABRIC impose compliance burden	Compliance	M	H	FABRIC read-only integration design avoids GxP validation burden on source systems. FABRIC itself follows Computer System Assurance (CSA) approach per FDA 2022 guidance for applicable analytical functions. Engage Baxter Compliance in Month 1 to agree on CSA scope.
Key talent shortage — knowledge engineers and semantic architects	Resourcing	M	H	TVE Group provides knowledge engineering as core competency for Phase 1-2. Parallel training of 2 Baxter internal staff in OWL/SHACL from Month 1 to build internal capability. Contingency: Stardog/GraphDB vendor professional services as backup resourcing.
Business scope creep from stakeholders	Program	H	M	Strict MVP definition per phase gate. Product Owners accountable for scope discipline within their application. Architecture Board reviews any scope addition with >5% effort impact before it enters backlog. PMO escalates to Program Director within 48 hours of first scope challenge.
AI model performance below user expectations in Phase 1	AI / Technical	M	M	Phase 1 AI use cases selected for high data quality domains and well-defined tasks. MTBH baseline established in Month 4 before any production user exposure. Circuit Breaker prevents low-confidence outputs from reaching users. Expectation-setting communication to users about AI-assisted (not AI-autonomous) framing.

9. Success Metrics & KPIs

KPI	Baseline	Phase 1 Target	Phase 2 Target	Phase 3 Target
R&D cross-BL formulation query time	Days (manual search)	<30 minutes	<5 minutes	Real-time (seconds)
Batch QC release decision support time	N/A (no system)	Batch Intelligence live	<4 hours (vs. 24+)	<1 hour
Data catalog coverage (%)	~10% estimated	40%	75%	95%
SHACL compliance rate (%)	N/A	Baseline established	>97%	>99.5%
Weekly active FABRIC users	0	20–30 (R&D pilot)	150–200	400+
Capability applications live (#)	0	2	6	8
Platform TCO vs. Palantir equivalent	Benchmark only	Infrastructure baseline set	\$6–10M savings demonstrated	\$6–13M/yr sustained savings
MTBH score (primary AI models)	N/A	Baseline defined per model	Within threshold 100% of window	Continuous monitoring; threshold met
Legacy systems decommissioned (#)	0	2–3 (tactical tools)	4–6	8–10+
Data lineage completeness (%)	~0%	40% (R&D domain)	80%	100%
Ontology concept coverage (%)	~0%	30% (R&D domain)	65%	85%
EU AI Act readiness score	N/A	Assessment complete	Remediation plan active	>85% compliant

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